

Title

Non-inferior detection of advanced retinopathy with risk-based variable screening intervals: outcomes from a large-scale population equivalence study

Abstract

Background: Risk-stratified, variable screening intervals might reduce healthcare burden in diabetes care while maintaining safety for retinopathy detection. **Methods:** 4534 individuals with diabetes were randomized to individualized risk-based screening intervals (6, 12, or 24 months; n=2097) or annual screening (n=2224). Screening risk categories were determined by a validated algorithm at baseline and updated after each visit. Primary outcome was attendance at first follow-up; secondary outcomes were sight-threatening retinopathy detection rates, quality-adjusted life-years, and incremental costs. **Results:** First visit attendance rates were equivalent between groups (individualized 83.6% vs. annual 84.7%; difference -1.0%, 95% CI -3.2 to 1.2). Sight-threatening retinopathy detection showed non-inferiority in the individualized group (1.4% vs. 1.7%; difference -0.3%, 95% CI -1.1 to 0.5). No serious adverse events were reported. Quality-adjusted life-year differences did not significantly differ from zero. Individualized screening reduced total appointments by 43.2% and generated incremental cost savings of £17.34 per person (95% CI £17.02-£17.67) from healthcare perspective and £23.11 (95% CI £22.73-£23.53) from societal perspective, representing 21% overall cost reduction. **Conclusions:** Risk-stratified, individualized screening intervals can be safely and cost-effectively implemented in established diabetic retinopathy screening programs.